KAUST Mathematics Competition (KMC): Regional Qualifier

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

KAUST Mathematics Competition (KMC): Regional Qualifier Practice Exercises

1 [KAUST KMC — Regional Qualifier] By grouping consecutive perfect-square intervals along $y = \sqrt{x}$ shown in the plot, compute the exact value of $\sum_{k=1}^{15} \lfloor \sqrt{k} \rfloor$.

a. 38

b. 14

c. 34 ✓

d. 31

For each integer $j \in \{1, 2, \ldots, 3\}$, $\lfloor \sqrt{k} \rfloor = j$ holds on $j^2 \le k \le (j+1)^2 - 1$, which contains $2j+1$ integers. Thus $\sum_{k=1}^{15} \lfloor \sqrt{k} \rfloor = \sum_{j=1}^{3} j(2j+1) = \frac{(3)(4)(4 \cdot 4 + 1)}{6} = 34$.

2 [KAUST KMC — Regional Qualifier] By grouping consecutive perfect-square intervals along $y = \sqrt{x}$ shown in the plot, compute the exact value of $\sum_{k=1}^{24} \lfloor \sqrt{k} \rfloor$.

a. 75

b. 70 ✓

c. 30

d. 66

For each integer $j \in \{1, 2, \ldots, 4\}$, $\lfloor \sqrt{k} \rfloor = j$ holds on $j^2 \le k \le (j+1)^2 - 1$, which contains $2j+1$ integers. Thus $\sum_{k=1}^{24} \lfloor \sqrt{k} \rfloor = \sum_{j=1}^{4} j(2j+1) = \frac{(4)(5)(4 \cdot 5 + 1)}{6} = 70$.

3 [KAUST KMC — Regional Qualifier] By grouping consecutive perfect-square intervals along $y = \sqrt{x}$ shown in the plot, compute the exact value of $\sum_{k=1}^{35} \lfloor \sqrt{k} \rfloor$.

a. 125 ✓

b. 131

c. 55

d. 120

For each integer $j \in \{1, 2, \ldots, 5\}$, $\lfloor \sqrt{k} \rfloor = j$ holds on $j^2 \le k \le (j+1)^2 - 1$, which contains $2j+1$ integers. Thus $\sum_{k=1}^{35} \lfloor \sqrt{k} \rfloor = \sum_{j=1}^{5} j(2j+1) = \frac{(5)(6)(4 \cdot 6 + 1)}{6} = 125$.

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4 [KAUST KMC — Regional Qualifier] By grouping consecutive perfect-square intervals along $y = \sqrt{x}$ shown in the plot, compute the exact value of $\sum_{k=1}^{48} \lfloor \sqrt{k} \rfloor$.

a. 210

b. 91

c. 197

d. 203 ✓

For each integer $j \in \{1, 2, \ldots, 6\}$, $\lfloor \sqrt{k} \rfloor = j$ holds on $j^2 \le k \le (j+1)^2 - 1$, which contains $2j + 1$ integers. Thus $\sum_{k=1}^{48} \lfloor \sqrt{k} \rfloor = \sum_{j=1}^{6} j(2j+1) = \frac{(6)(7)(4 \cdot 7 + 1)}{6} = 203$.

5 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a **monic** polynomial of degree 3 such that $P(k) = 2k$ for every integer $k \in \{1, 2, \ldots, 3\}$. Find the exact value of $P(4)$ (note the classic linear-extrapolation trap 8!).

a. 8

b. 6

c. 14 ✓

d. 12

Define the auxiliary polynomial $Q(x) = P(x) - 2x$. Since P is monic of degree $3 \ge 3$, $Q(x)$ is also monic of degree 3 and vanishes at $1, 2, \ldots, 3$. By the Factor Theorem, $P(x) - 2x = \prod_{k=1}^{3} (x - k)$. Substituting $x = 4$ yields $P(4) = 6 + 2(4) = 14$.

6 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a **monic** polynomial of degree 4 such that $P(k) = 2k$ for every integer $k \in \{1, 2, \ldots, 4\}$. Find the exact value of $P(6)$ (note the classic linear-extrapolation trap 12!).

a. 12

b. 132 ✓

c. 120

d. 32

Define the auxiliary polynomial $Q(x) = P(x) - 2x$. Since P is monic of degree $4 \ge 3$, $Q(x)$ is also monic of degree 4 and vanishes at $1, 2, \ldots, 4$. By the Factor Theorem, $P(x) - 2x = \prod_{k=1}^{4} (x - k)$. Substituting $x = 6$ yields $P(6) = 120 + 2(6) = 132$.

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7 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a **monic** polynomial of degree 5 such that $P(k) = 2k$ for every integer $k \in \{1, 2, \ldots, 5\}$. Find the exact value of $P(6)$ (note the classic linear-extrapolation trap 12!).

a. 132 ✓

b. 12

c. 120

d. 130

Define the auxiliary polynomial $Q(x) = P(x) - 2x$. Since P is monic of degree $5 \geq 3$, $Q(x)$ is also monic of degree 5 and vanishes at $1, 2, \ldots, 5$. By the Factor Theorem, $P(x) - 2x = \prod_{k=1}^{5} (x - k)$. Substituting $x = 6$ yields $P(6) = 120 + 2(6) = 132$.

8 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a **monic** polynomial of degree 3 such that $P(k) = 3k$ for every integer $k \in \{1, 2, \ldots, 3\}$. Find the exact value of $P(5)$ (note the classic linear-extrapolation trap 15!).

a. 15

b. 24

c. 40

d. 39 ✓

Define the auxiliary polynomial $Q(x) = P(x) - 3x$. Since P is monic of degree $3 \geq 3$, $Q(x)$ is also monic of degree 3 and vanishes at $1, 2, \ldots, 3$. By the Factor Theorem, $P(x) - 3x = \prod_{k=1}^{3} (x - k)$. Substituting $x = 5$ yields $P(5) = 24 + 3(5) = 39$.

9 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a polynomial of degree **at most** 2 satisfying $P(k) = \frac{24}{k}$ for $k = 1, 2, \ldots, 3$. Compute the exact value of $P(4)$.

a. 6

b. 16

c. 12 ✓

d. 13

Clear denominators by setting $Q(x) = xP(x) - 24$, which has degree at most 3 and roots $1, 2, \ldots, 3$. Thus $Q(x) = A \prod_{k=1}^{3} (x - k)$. Evaluating at the hidden anchor $x = 0$ gives $-24 = A(-1)^3(3)! \Rightarrow A = 4$. At $x = 4$, $4P(4) - 24 = 24 \Rightarrow P(4) = 12$.

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10 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a polynomial of degree **at most** 3 satisfying $P(k) = \frac{120}{k}$ for $k = 1, 2, \ldots, 4$. Compute the exact value of $P(5)$.

a. 24

b. 0 ✓

c. 5

d. 48

Clear denominators by setting $Q(x) = xP(x) - 120$, which has degree at most 4 and roots $1, 2, \ldots, 4$. Thus $Q(x) = A\prod_{k=1}^{4}(x-k)$. Evaluating at the hidden anchor $x = 0$ gives $-120 = A(-1)^4(4)! \Rightarrow A = -5$. At $x = 5$, $5P(5) - 120 = -120 \Rightarrow P(5) = 0$.

11 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a polynomial of degree **at most** 4 satisfying $P(k) = \frac{720}{k}$ for $k = 1, 2, \ldots, 5$. Compute the exact value of $P(6)$.

a. 240 ✓

b. 120

c. 246

d. 241

Clear denominators by setting $Q(x) = xP(x) - 720$, which has degree at most 5 and roots $1, 2, \ldots, 5$. Thus $Q(x) = A\prod_{k=1}^{5}(x-k)$. Evaluating at the hidden anchor $x = 0$ gives $-720 = A(-1)^5(5)! \Rightarrow A = 6$. At $x = 6$, $6P(6) - 720 = 720 \Rightarrow P(6) = 240$.

12 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a polynomial of degree **at most** 5 satisfying $P(k) = \frac{5040}{k}$ for $k = 1, 2, \ldots, 6$. Compute the exact value of $P(7)$.

a. 720

b. 7

c. 1440

d. 0 ✓

Clear denominators by setting $Q(x) = xP(x) - 5040$, which has degree at most 6 and roots $1, 2, \ldots, 6$. Thus $Q(x) = A\prod_{k=1}^{6}(x-k)$. Evaluating at the hidden anchor $x = 0$ gives $-5040 = A(-1)^6(6)! \Rightarrow A = -7$. At $x = 7$, $7P(7) - 5040 = -5040 \Rightarrow P(7) = 0$.

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